

Existing Project mein React Add Karna

Is page ka main idea ye hai ki agar tumhare paas pehle se koi website ya web application hai, toh use **poora rewrite karne ki zarurat nahi hai**. Tum apne existing HTML, CSS, JavaScript ya backend project ke andar React ko gradually add kar sakte ho.

React ko mainly do tarikon se integrate kiya ja sakta hai:

1. Website ke kisi **poore subroute/section** ke liye
 2. Existing page ke andar sirf kuch **interactive components** ke liye
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Pehli requirement: Node.js

Local development ke liye tumhare system mein **Node.js install** hona chahiye.

Node.js isliye zaroori hai kyunki React development mein commonly use hone wale tools, jaise:

- npm packages
- Vite
- Babel
- JavaScript build tools

Node.js environment par depend karte hain.

React ko simple HTML page ya online environment mein try kiya ja sakta hai, lekin practical project development ke liye Node.js ki zarurat padegi.

Approach 1: Existing website ke poore subroute mein React

Maan lo tumhari existing website hai:

`1 example.com`

Ye website Rails ya kisi doosri server-side technology par bani hui hai. Ab tum chahte ho ki iske andar:

`1 example.com/some-app/`

se start hone wale saare routes React se banaye jaayen.

Example:

```
1 example.com/some-app/  
2 example.com/some-app/profile  
3 example.com/some-app/settings
```

Is case mein /some-app/ website ka React-controlled section hoga.

Isko setup karne ke steps

1. React-based framework se React application banao

React part ko kisi React-based framework ke through build karo, jaise Next.js ya Gatsby.

Framework use karne ka benefit ye hai ki routing, optimization aur build configuration jaise important features pehle se available hote hain.

2. Base path configure karo

Framework configuration mein /some-app ko **base path** set karo.

Base path framework ko batata hai ki React application website ke root / par nahi, balki /some-app/ ke andar run hogi.

Conceptually:

```
1 Main website: /  
2 React application: /some-app/
```

Next.js aur Gatsby dono base path ya path prefix configure karne ka option dete hain.

3. Server ya proxy configure karo

Apne existing server ya proxy ko configure karo taaki:

```
1 /some-app/
```

ke under aane wali requests React application handle kare.

Baaki routes existing application handle karti rahegi.

```
1 example.com/about      → Existing application  
2 example.com/contact    → Existing application  
3 example.com/some-app/  → React application
```

JavaScript server par run karna compulsory nahi hai

Kaafi React frameworks full-stack hote hain, jiska matlab hai ki React application server-side functionality bhi use kar sakti hai.

Lekin agar tum:

- Server par JavaScript run nahi kar sakte
- JavaScript server use nahi karna chahte

- Sirf static files deploy karna chahte ho

toh React application ka exported HTML, CSS aur JavaScript output /some-app/ par serve kar sakte ho.

Is approach mein React app ka static build server par host kiya jaata hai.

Approach 2: Existing page ke ek part mein React

Maan lo tumhara existing page Rails, Backbone, normal HTML ya kisi aur technology se bana hua hai. Tum poore page ko React mein convert nahi karna chahte, balki page ke andar sirf ek interactive component add karna chahte ho.

Examples:

- Navigation bar
- Search box
- Like button
- Form
- Dropdown
- Shopping cart
- Interactive dashboard section

Ye React integrate karne ka common approach hai. Page ke existing parts same rehte hain aur selected HTML elements ke andar React components render kiye jaate hain.

Iske do main steps hain:

1. Modular JavaScript environment setup karna
 2. Required location par React component render karna
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Step 1: Modular JavaScript Environment Setup Karna

Modular JavaScript environment tumhe code ko multiple files mein divide karne deta hai.

Example:

```
1  src/  
2  |— index.js  
3  |— NavigationBar.jsx  
4  |— SearchBox.jsx  
5  |— Button.jsx
```

Isse tum:

- import aur export use kar sakte ho
- JSX syntax likh sakte ho
- npm packages install kar sakte ho
- React components ko separate files mein organize kar sakte ho

Tumhara setup tumhare existing project par depend karega.

Case 1: Project mein already import use ho raha hai

Agar project ki JavaScript files mein pehle se aisa code hai:

```
1 import something from './something.js';
```

toh pehle check karo ki JSX syntax support ho raha hai ya nahi.

Example:

```
1 <div />
```

Agar is code se syntax error milta hai, toh possible hai ki existing build setup JSX ko understand nahi karta.

Aise case mein JavaScript code ko **Babel** se transform karna pad sakta hai aur JSX support ke liye **Babel React preset** enable karna hoga.

Babel JSX jaise syntax ko browser-compatible JavaScript mein convert karta hai.

Case 2: Project mein module/build setup nahi hai

Agar tumhara JavaScript code ek hi file mein hai aur project import/export use nahi karta, toh page ke according **Vite** setup karne ki recommendation di gayi hai.

Vite tumhe modern development environment deta hai, including:

- JavaScript modules
- JSX support
- Development server
- npm package integration
- Production build

Vite community Rails, Django aur Laravel jaise backend frameworks ke liye integrations maintain karti hai. Agar tumhara backend listed nahi hai, toh Vite ko manually backend ke saath integrate kiya ja sakta hai.

React Packages Install Karna

Apne project folder ke terminal mein run karo:

```
1 npm install react react-dom
```

Is command se do packages install honge:

- react: React components aur React functionality ke liye
 - react-dom: React components ko browser DOM mein render karne ke liye
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Initial React Test

Main JavaScript file usually inmein se koi ho sakti hai:

```
1 index.js
2 main.js
3 app.js
```

Page ke example mein pehle existing body content remove karke ek naya element create kiya gaya hai:

```
1 import { createRoot } from 'react-dom/client';
2
3 // Existing HTML content hata do
4 document.body.innerHTML = '<div id="app"></div>';
5
6 // React component render karo
7 const root = createRoot(document.getElementById('app'));
8 root.render(<h1>Hello, world</h1>);
```

Code ka breakdown

createRoot import karna

```
1 import { createRoot } from 'react-dom/client';
```

createRoot React ko kisi HTML element ke andar render karne ke liye use hota hai.

Body content replace karna

```
1 document.body.innerHTML = '<div id="app"></div>';
```

Ye poori existing body ko hata kar ek naya div bana raha hai:

HTML element find karna

```
1 document.getElementById('app')
```

Ye id="app" wale element ko DOM se find karta hai.

React root create karna

```
1 const root = createRoot(document.getElementById('app'));
```

React ko bataya ja raha hai ki uska content app element ke andar render hoga.

React content render karna

```
1 root.render(<h1>Hello, world</h1>);
```

React <h1> ko app element ke andar render karega.

Agar page ka entire content replace hokar **Hello, world!** show hota hai, iska matlab React setup kaam kar raha hai.

Ye code mainly setup test karne ke liye hai. Actual integration mein existing HTML ko delete nahi karna hai.

Step 2: Page par kahin bhi React component render karna

React setup verify karne ke baad ye test code delete karna hai:

```
1 document.body.innerHTML = '<div id="app"></div>';
```

Kyunki ye existing page content ko remove kar raha tha.

Uske badle existing HTML mein us jagah ek unique id add karo jahan React component show karna hai.

Example:

Yahan:

React component ka mounting point hai.

React component banana

```
1 import { createRoot } from 'react-dom/client';
2
3 function NavigationBar() {
4   return <h1>Hello from React!</h1>;
5 }
6
```

```
7 const domNode = document.getElementById('navigation');
8 const root = createRoot(domNode);
9 root.render(<NavigationBar />);
```

Is code ka flow

Component define hua

```
1 function NavigationBar() {
2   return <h1>Hello from React!</h1>;
3 }
```

NavigationBar ek React component hai jo <h1> return kar raha hai.

Target HTML element find hua

```
1 const domNode = document.getElementById('navigation');
```

Browser existing HTML ke andar ye element find karega:

React root create hua

```
1 const root = createRoot(domNode);
```

Ab React ko navigation element control karne ke liye diya gaya.

Component render hua

```
1 root.render(<NavigationBar />);
```

Result roughly aisa dikhega:

Existing HTML preserve kaise hota hai?

Original page:

React sirf is element ke andar render ho raha hai:

Isliye dono normal HTML paragraphs preserve rahenge. React poore page ko control nahi kar raha, sirf navigation section ko control kar raha hai.

Conceptually:

```
1 Normal HTML
2   ↓
3 React-controlled navigation
4   ↓
5 Normal HTML
```

Isi wajah se existing application mein React incrementally add kiya ja sakta hai.

Multiple Places Par React Render Karna

Page mein multiple mounting points bhi ho sakte hain:

Har location ke liye appropriate React component render kiya ja sakta hai. Lekin page ke provided example mein specifically ek `NavigationBar` component render karke basic integration samjhaya gaya hai.

Incremental React Adoption

Existing project mein React adopt karte waqt recommended pattern hai:

1. Ek small interactive component se start karo
2. Uske aas-paas ke related sections React mein convert karo
3. Dheere-dheere React-controlled area ko expand karo
4. Agar eventually poora page React mein convert ho jaaye, toh React framework par migrate karne par consider karo

Example progression:

1	Button
2	↓
3	Form
4	↓
5	Complete section
6	↓
7	Complete page
8	↓
9	React-based application

Is approach ka benefit ye hai ki tumhe existing application ek saath rewrite nahi karni padti.

React Native ko Existing Mobile App mein Add Karna

React Native ko bhi existing native mobile application mein incrementally integrate kiya ja sakta hai.

Supported existing apps mein include hain:

- Android app written in Java or Kotlin

- iOS app written in Objective-C or Swift

Poore native app ko rewrite karne ke badle tum existing application mein ek React Native screen add kar sakte ho.

Simple Decision Guide

Poora /some-app/ section React mein banana hai

React-based framework use karo, /some-app ko base path banao aur server/proxy routing configure karo.

Existing page mein ek component add karna hai

HTML mein unique id add karo aur `createRoot()` ke through us element mein React component render karo.

Existing project JSX support nahi karta

Existing compilation setup check karo. Zarurat ho toh Babel React preset use karo.

Project mein modern JavaScript setup nahi hai

Vite ke through modular environment setup karo.

Existing native mobile app mein React add karna hai

React Native integration guide follow karke individual React Native screen add karo.

Sabse Important Takeaway

React **all-or-nothing technology nahi hai**. Tumhe React use karne ke liye apna existing project delete ya rewrite karna zaroori nahi hai.

Tum React ko:

- Ek button mein
- Ek navigation bar mein
- Ek page section mein
- Ek complete subroute mein
- Ya existing native mobile app ki ek screen mein

gradually introduce kar sakte ho.

External Links Mentioned on the Page

Mozilla/MDN ke links aur React website ke internal links ko exclude karke page par ye external resources diye gaye hain:

- <https://nodejs.org/en/>
- [Next.js basePath configuration](#)
- <https://www.gatsbyjs.com/docs/how-to/previews-deploys-hosting/path-prefix/>
- [Next.js static HTML export](#)
- <https://www.npmjs.com/>
- <https://babeljs.io/setup>
- <https://babeljs.io/docs/babel-preset-react>
- [Vite documentation](#)
- [Vite backend framework integrations](#)
- [Vite backend integration guide](#)
- [Vite community chat](#)
- [React Native](#)
- <https://reactnative.dev/docs/integration-with-existing-apps>